(Please write your Enrollment Number) Enrollment No. 01001042022

MID-TERM EXAMINATION

(Course Name B.Tech MAE/DMAM) (Semester 3rd)

(October, 2023) OFF LINE mode

Subject Code: BMA 203	**Subject: Strength of Materials**
Time : 1 ½Hours	**Maximum Marks : 30**
Note: Q. 1 is compulsory. Assume all variables required during answering of questions	

Q1		**(2.5*4)**	
	(a) State stress at a point		
	(b) Define principal plane		
	(c) Establish SF & BM relationship of uniformly distributed load		
	(d) Enlist assumptions of simple theory of bending		
Q2	**(Attempt any Two Parts) UNIT-1**	**(5,5)**	
	(a) Establish relationship of elongation of uniformly tapered rectangular bar subjected to tensile force		
	(b) Prove **E=3K(1-2v)**		
	(c) Find max shear stress for a stress tensor σ_{xx}= 15 MPa, σ_{yy} = 10 MPa & τ_{xy}= 5 MPa		
Q3	**(Attempt any Two Parts) UNIT-2**	**(5,5)**	
	(a) Draw SF and BM of a simply supported beam subjected to distributed load $w=w_0(1+x)$		
	(b) Derive simple theory of bending formula		
	(c) Determine maximum deflection of a simply supported beam subjected to eccentric load at a distance "a" from left hand side support		